Mechanical CAD Portfolio

3D Modelling | 2D Drawings | Assemblies
Design for Manufacturing



About Me

I am a Mechanical CAD designer with hands-on experience in 3D modelling, 2D technical drawings, and assembly design. I use industry-standard CAD tools to develop accurate and manufacturable designs that meet functional and production requirements.

Core Skills

- 3D Part & Assembly Modelling
- 2D Technical Drawings (GD&T)
- Sheet Metal Design
- Design for Manufacturability (DFM)
- Assembly & Exploded Views
- CAD File Management & Revision Control

Software Proficiency

SolidWorks | Inventor | AutoCAD | Fusion 360

CONTACT

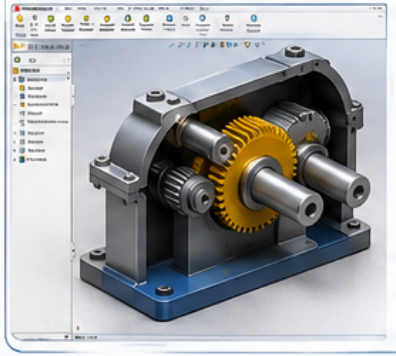
Portfolio owner
Potter Christina Joy

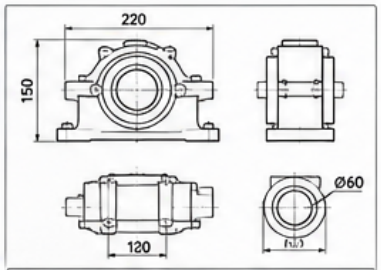
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Featured Projects

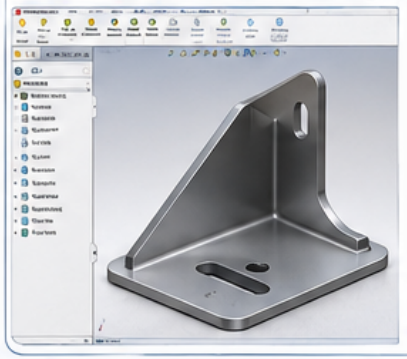
1. Gearbox Assembly

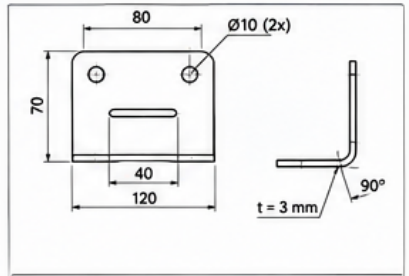




GEARBOX HOUSING			
SCALE: 1:2	MAT: Cast Iron	DRG NO: GBX-001	REV: A

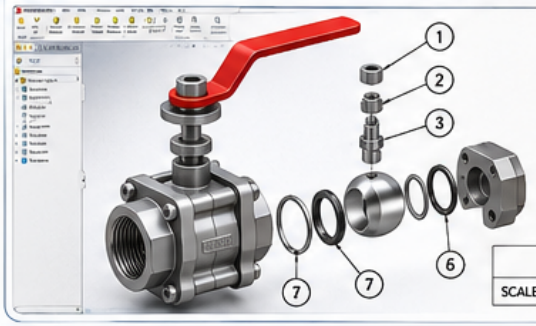
2. Sheet Metal Bracket





SHEET METAL BRACKET			
SCALE: 1:1	MAT: Mild Steel	DRG NO: BRK-002	REV: A

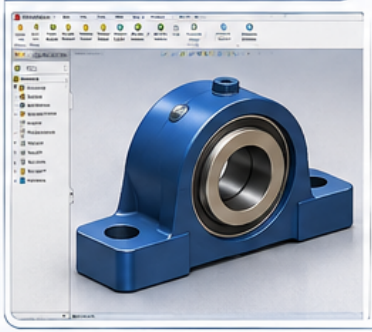
3. Valve Assembly (Exploded View)

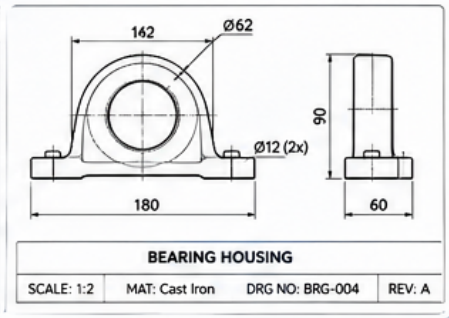


PARTS LIST		
ITEM	DESCRIPTION	QTY
1	Handle	1
2	Nut	1
3	Stem	1
4	Ball	1
5	Seat Ring	2
6	Body	1
7	End Cap	1

BALL VALVE ASSEMBLY		
SCALE: 1:2	DRG NO: VLV-001	REV: A

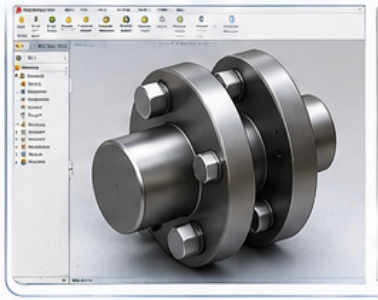
4. Bearing Housing

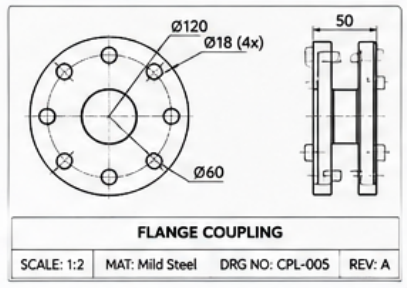




BEARING HOUSING			
SCALE: 1:2	MAT: Cast Iron	DRG NO: BRG-004	REV: A

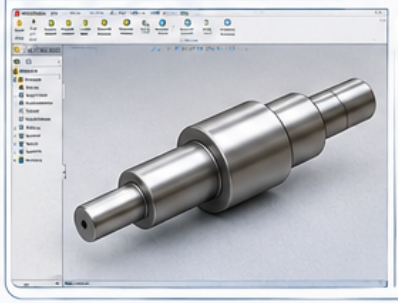
5. Flange Coupling

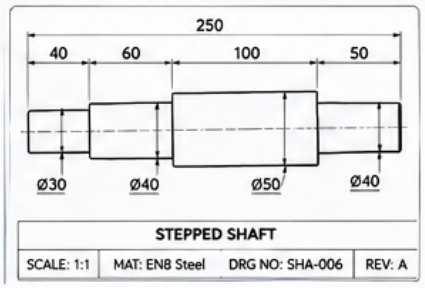




FLANGE COUPLING			
SCALE: 1:2	MAT: Mild Steel	DRG NO: CPL-005	REV: A

6. Stepped Shaft

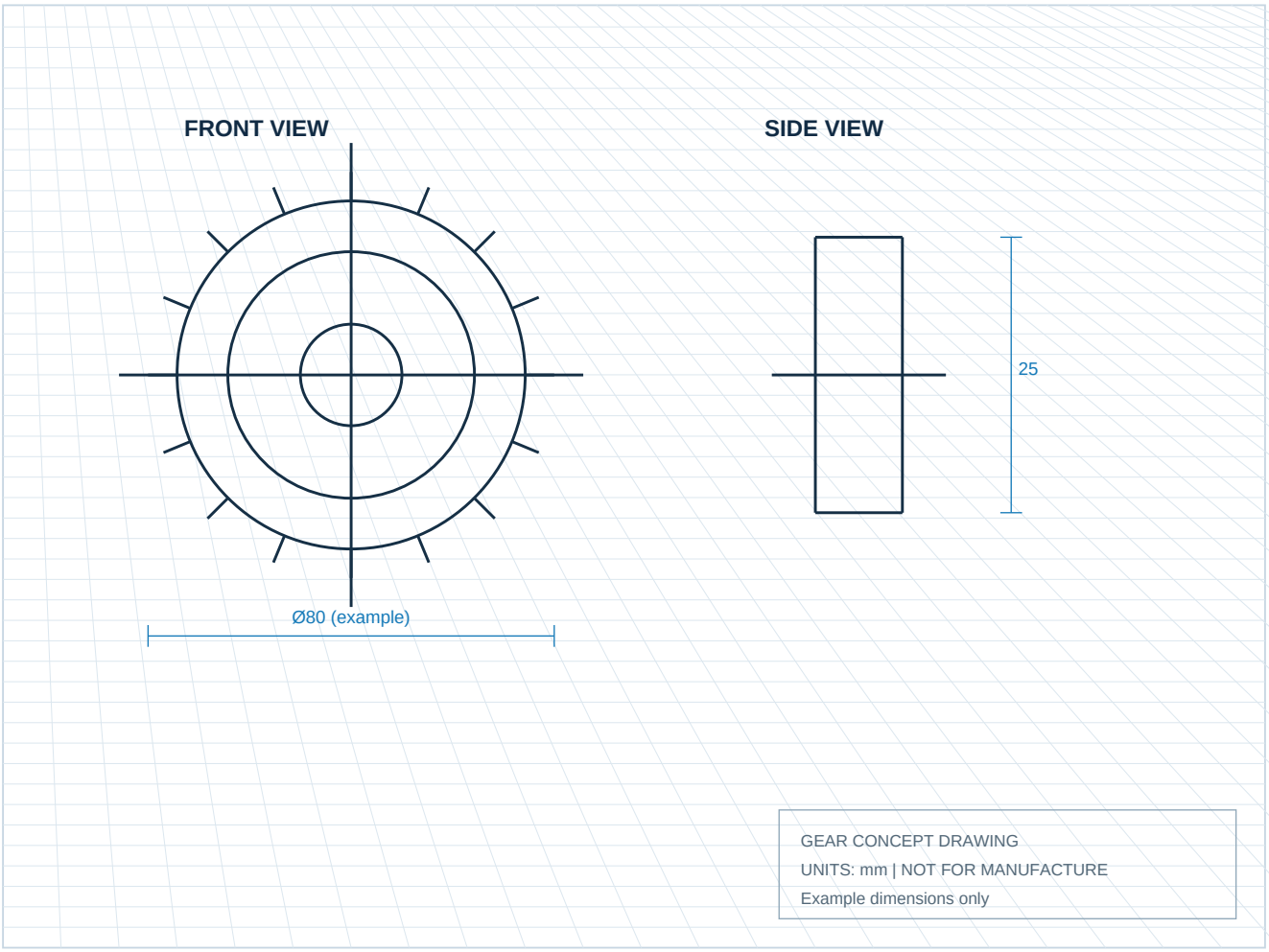




STEPPED SHAFT			
SCALE: 1:1	MAT: EN8 Steel	DRG NO: SHA-006	REV: A

Additional 2D CAD Drawing — Spur Gear

A simple drawing layout with front and side views, centre lines, and example size callouts.

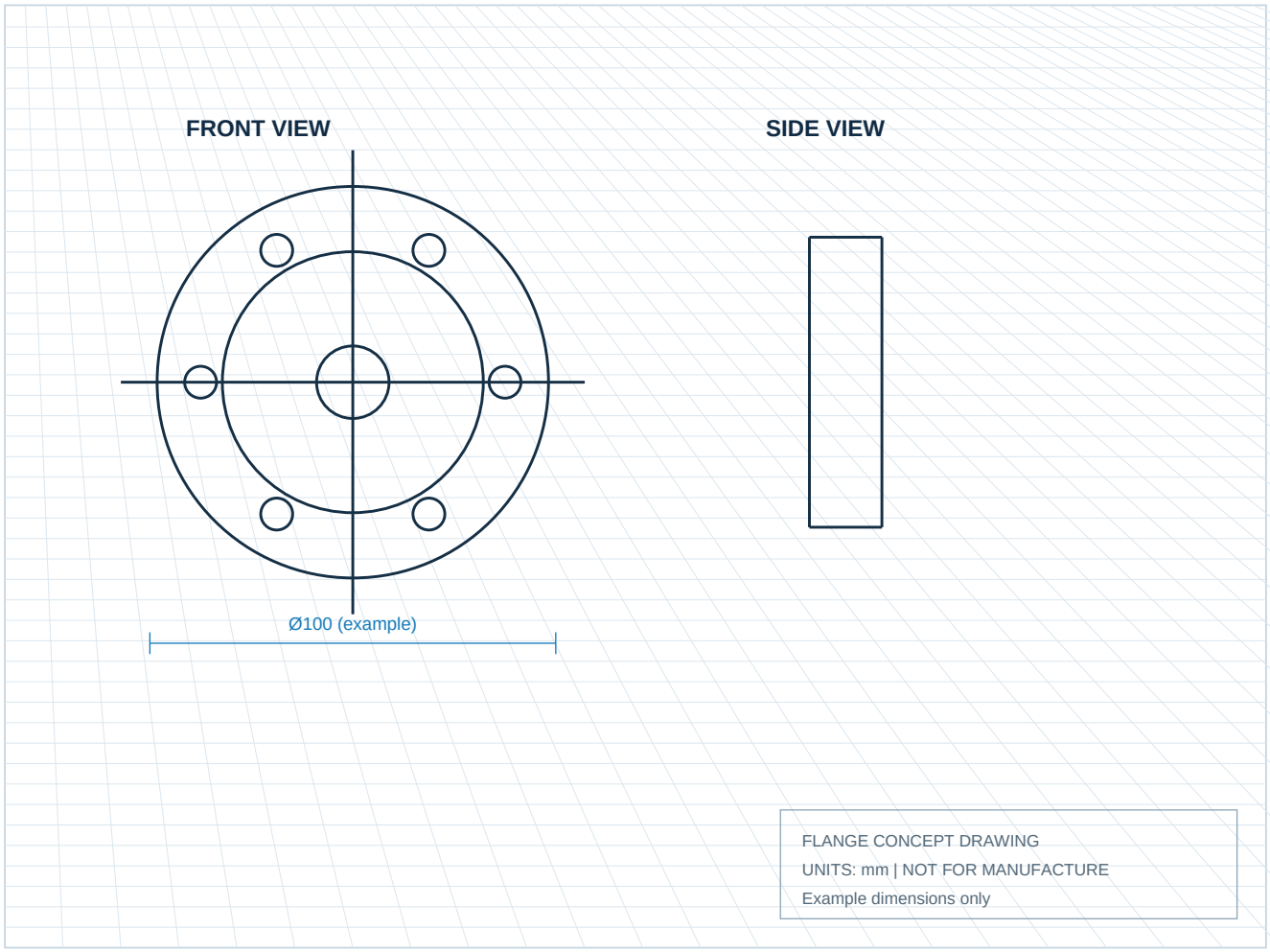


Drawing notes: The front view shows the bore and gear outline. The side view gives an example thickness. A full manufacturing drawing would need tooth data, tolerances, material, and a suitable standard.

These are illustrative practice sketches. Dimensions are examples only and must be checked against a real design brief before use.

Additional 2D CAD Drawing — Flange

A simple flange layout with a centre bore and evenly spaced bolt holes.

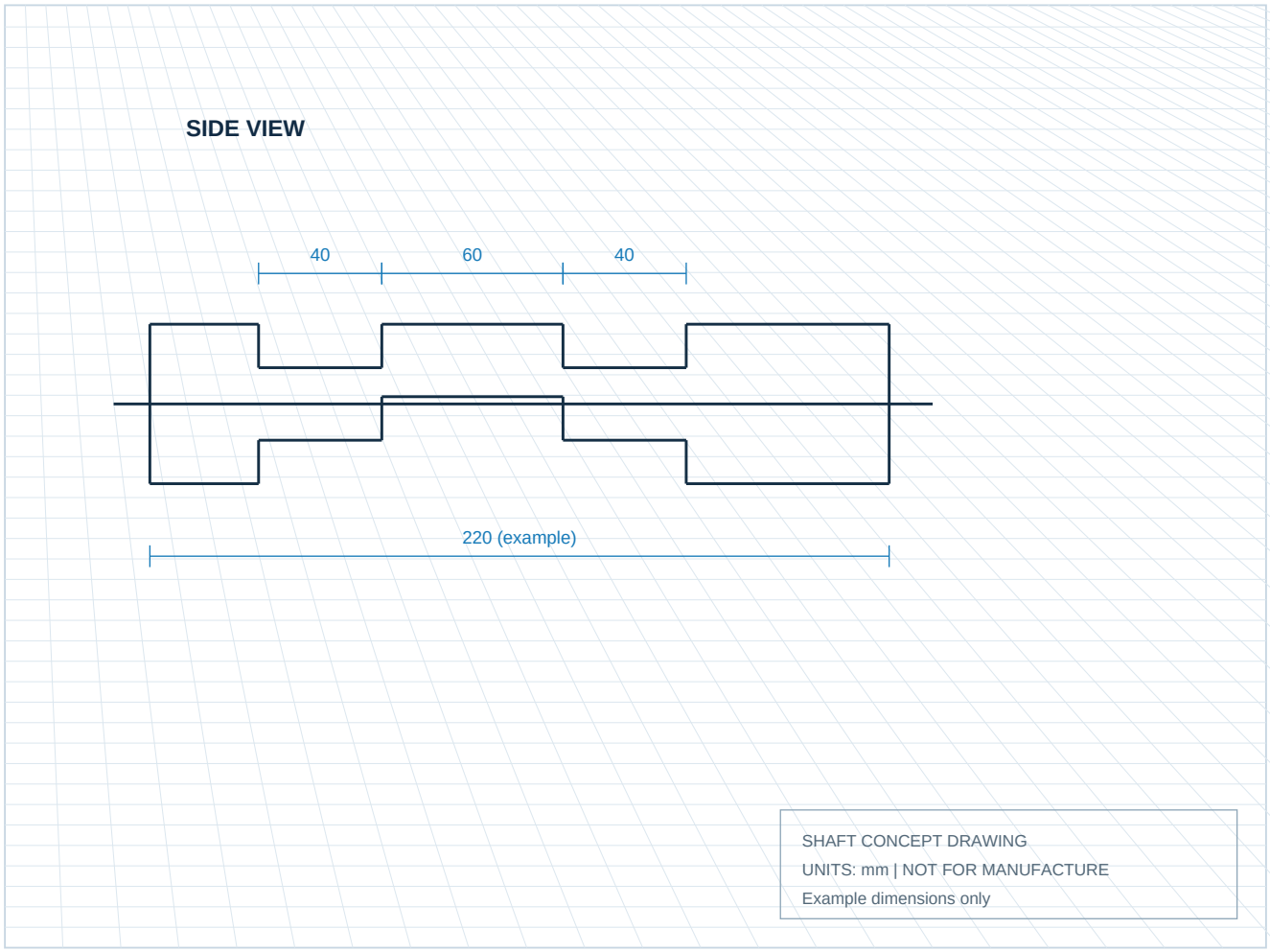


Drawing notes: The drawing demonstrates a circular part layout and bolt-hole pattern. Confirm bolt-circle size, hole diameter, thickness, and sealing needs for any real application.

These are illustrative practice sketches. Dimensions are examples only and must be checked against a real design brief before use.

Additional 2D CAD Drawing — Stepped Shaft

A stepped shaft side view with example section lengths and an overall length.

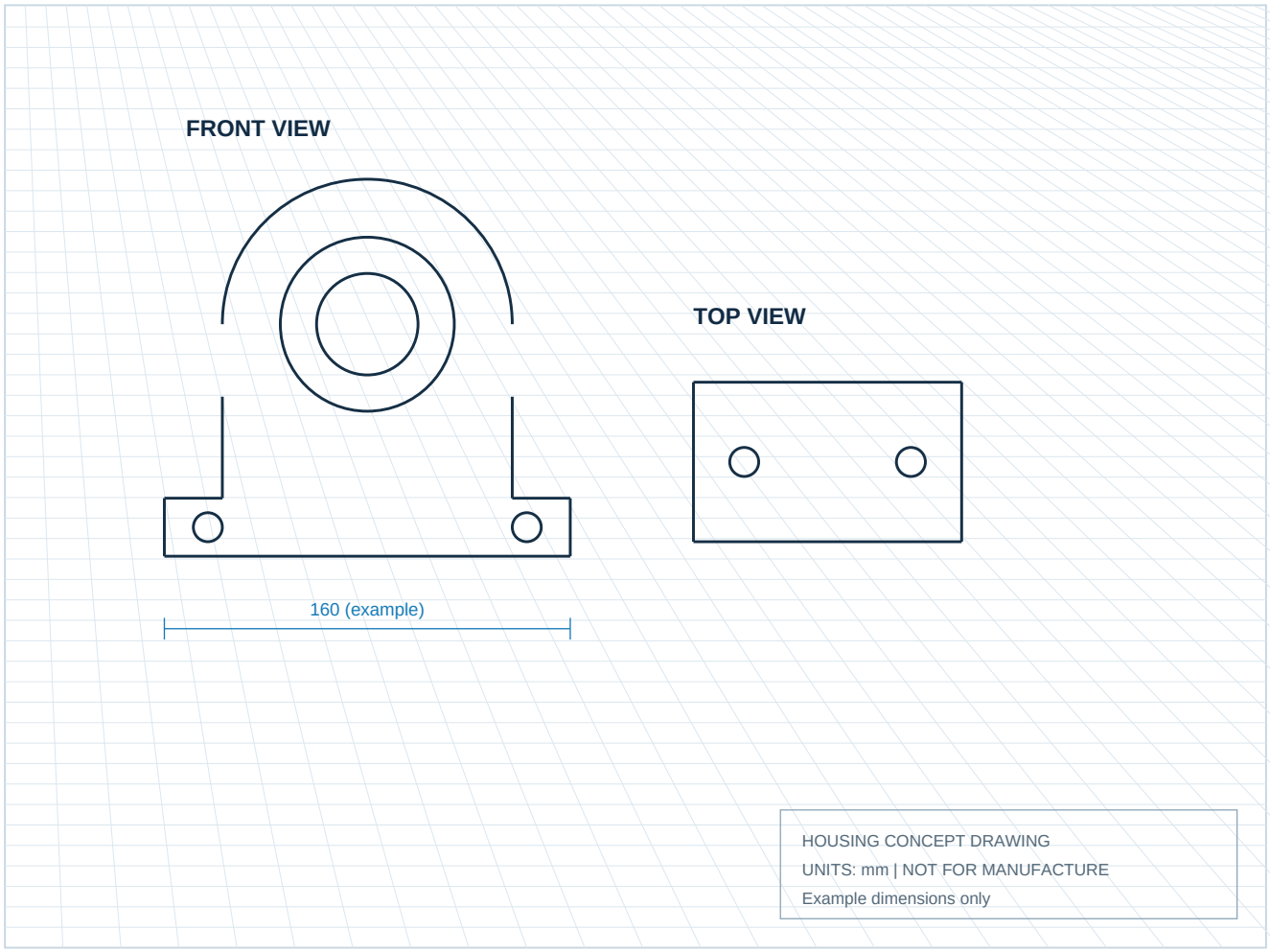


Drawing notes: The drawing shows changes in shaft diameter and section length. A real shaft drawing should include all diameters, fits, surface finish, material, and tolerances.

These are illustrative practice sketches. Dimensions are examples only and must be checked against a real design brief before use.

Additional 2D CAD Drawing — Bearing Housing

Front and top views of a simple bearing housing concept.



Drawing notes: The drawing shows a housing outline, bearing seat, and mounting holes. A real design needs bearing dimensions, fit details, wall thickness checks, and load review.

These are illustrative practice sketches. Dimensions are examples only and must be checked against a real design brief before use.